

Abstract

This study extends the UBP–RLS (Universal Binary Principle – Radial Layer Spiral) fusion framework from 5×10^4 to 10^6 cells, achieving 1° angular resolution (360 sectors) compared to the previous 10° (36 sectors). We investigate whether the near-perfect angular correlations ($r > 0.99$) reported in Phase XIII persist at finer scales, and explore three new research directions: (1) the spatial distribution of prime k -tuples on the RLS and their relationship to the Hardy–Littlewood conjectures, (2) the angular clustering behaviour of primes in short intervals $[x, x + \sqrt{x}]$, and (3) the three-dimensional structure of the RLS when a temporal dimension (layer index, $\log n$, or Lucas–Lehmer iteration depth) is introduced as a third axis.

Key findings include: (i) UBP–prime angular correlations remain robust at 1° resolution ($r_{\text{RotSC}} = +0.9696$, $r_{\text{Anchor}} = +0.9686$), though NRCI_{GL} degrades from -0.93 to -0.72 ; (ii) the RLS exhibits strong 4-fold circular symmetry ($r = 0.984$ at 90° lag); (iii) all k -tuple densities track prime density ($r > 0.99$), but the *per-prime* k -tuple rate is uniform across angular sectors, implying the RLS captures prime *density* but not k -tuple *correlations* beyond the trivial effect of more primes in certain sectors; (iv) twin primes reject uniformity across sectors ($\chi^2 = 1951$, $\text{df} = 35$); (v) the 3D voxel representation reveals 21% more prime-density variance than the 2D projection, suggesting the temporal dimension encodes genuine structural information; and (vi) primes $p \equiv 1 \pmod{4}$ and $p \equiv 3 \pmod{4}$ exhibit fundamentally different angular structures (cross-correlation $r = -0.40$).

1 Introduction

The Radial Layer Spiral (RLS), introduced by Waluś [1], maps natural numbers onto a two-dimensional lattice by decomposing each number as a point (i, j) on circles defined by $i^2 + j^2 = m$, ordered by increasing m and, within each circle, by angular position $\theta = \arctan(j/i)$. This construction is deeply connected to the theory of Gaussian integers: primes $p \equiv 1 \pmod{4}$ lie on RLS layers (they admit representation as sums of two squares), while primes $p \equiv 3 \pmod{4}$ do not (they are assigned positions by the sequential numbering, but their layer m is never a perfect sum of two squares).

In Phase XIII of the UBP–RLS fusion study [2], we demonstrated that projecting UBP (Universal Binary Principle) dynamic metrics — including Rotation Sign Changes (RSC), Anchor Distance, Mean Angle, and NRCI (Golay–Leech Normalised Residual Correction Integrity) — onto the RLS angular coordinate system yields near-perfect correlation with prime density across 36 angular sectors. The strongest correlations were: Mean Angle $r = +0.9906$, Anchor Distance $r = +0.9901$, and RSC $r = +0.9894$. This finding suggested that the discrete 24-dimensional UBP topological signals and the continuous 2-dimensional RLS geometric structure are two views of the same underlying phenomenon.

The present study extends this analysis in four directions, each addressing a fundamental question about the UBP–RLS relationship:

- (i) **Scale and resolution.** Does the $r > 0.99$ correlation survive at 1° angular resolution on a 10^6 -cell grid? The previous study used 10° sectors and 5×10^4 cells; finer resolution tests whether the signal is a coarse-grained artefact or a genuine fine-structure property.
- (ii) **Higher-order prime distributions.** Do prime k -tuples (twins, cousins, sexy primes, triplets, quadruplets) exhibit angular clustering on the RLS beyond what is expected from their tracking of prime density? Furthermore, do the angular distributions satisfy or violate the Hardy–Littlewood prime k -tuple conjectures?
- (iii) **Short-interval angular structure.** For primes within short intervals $[x, x + \sqrt{x}]$, which are classically expected to be uniformly distributed, does the RLS angular coordinate reveal unexpected clustering at small scales?

- (iv) **Three-dimensional temporal structure.** The RLS is inherently a 2D projection of what may be a higher-dimensional object. When a temporal axis (layer index, $\log n$, or LL iteration depth) is introduced as the third dimension, does the 3D structure reveal information invisible in the 2D projection? This addresses the hypothesis that the 2D RLS “forces” numbers into a specific shape, and the 3D version may show the true underlying geometry.

2 Methodology

2.1 Extended RLS Construction

Definition 2.1 (RLS Layer). For a non-negative integer m , the *RLS layer* $\mathcal{L}_m$ is the set of integer pairs (i, j) satisfying $i^2 + j^2 = m$, ordered by angular position $\theta = \arctan(j/i) \in [0, 2\pi)$. Natural numbers are assigned sequentially across layers in order of increasing m , and within each layer in order of increasing θ .

The Phase XIV grid was constructed for $N = 10^6$ cells, yielding 72,337 layers with $m_{\max} = 318,298$ and $r_{\max} = 564.2$. Primality was determined using the Miller–Rabin test with $k = 12$ deterministic witnesses. The resulting grid contains 78,498 primes (7.85%), consistent with the expected $\sim 1/\ln(10^6) \approx 7.24\%$ given the non-uniform distribution across the RLS.

2.2 UBP Metrics

The following UBP dynamic metrics were computed for a stratified random sample of 200,000 cells (every 5th cell):

- **NRCI_{GL}**: Golay–Leech Normalised Residual Correction Integrity, computed via Gray encoding $\rightarrow$ Golay [24, 12, 8] decode $\rightarrow$ Leech lattice NRCI.
- **RSC** (Rotation Sign Changes): Number of direction reversals in the 4D prime-residue fingerprint trajectory during Lucas–Lehmer iteration $s_{k+1} = s_k^2 - 2 \pmod{n}$.
- **Anchor Distance**: Hamming distance from the Golay-decoded codeword to the nearest valid Golay codeword.
- **Mean Angle**: Average angular deflection between consecutive LL trajectory steps.
- **Hamming Weight**: Weight of the Golay-encoded codeword.

2.3 Angular Sector Analysis

The 360° circle was divided into either 36 sectors of 10° each (for comparison with Phase XIII) or 360 sectors of 1° each (the new fine-resolution analysis). For each sector S_k , we computed the prime density $\rho_k = |\{n \in S_k : n \text{ is prime}\}|/|S_k|$ and the mean of each UBP metric across cells in S_k . The Pearson correlation $r(\rho, M)$ between prime density and each metric M was then evaluated across sectors.

2.4 Prime K -Tuple Identification

Six types of prime k -tuples were identified by scanning the full 10^6 -cell grid: twins $(p, p + 2)$, cousins $(p, p + 4)$, sexy primes $(p, p + 6)$, two types of prime triplets $(p, p + 2, p + 6)$ and $(p, p + 4, p + 6)$, and prime quadruplets $(p, p + 2, p + 6, p + 8)$. For each tuple type, we computed the angular distribution across 36 sectors and the Rayleigh test statistic R for circular uniformity.

2.5 Hardy–Littlewood Conjecture Testing

For each admissible k -tuple pattern $\mathcal{H} = (h_1, \dots, h_k)$, the first Hardy–Littlewood conjecture predicts

$$\pi_{\mathcal{H}}(N) \sim C(\mathcal{H}) \int_2^N \frac{dt}{(\log t)^k}, \quad (1)$$

where the singular series constant $C(\mathcal{H})$ is given by

$$C(\mathcal{H}) = \prod_p \left(1 - \frac{1}{p}\right)^{-k} \left(1 - \frac{\nu_p(\mathcal{H})}{p}\right), \quad (2)$$

with $\nu_p(\mathcal{H})$ denoting the number of distinct residue classes occupied by $\mathcal{H}$ modulo p . We computed $C(\mathcal{H})$ for each pattern and compared the predicted count to the observed count in the 10^6 -cell RLS grid.

2.6 Three-Dimensional RLS

The RLS was extended to three dimensions by introducing a temporal coordinate z for each cell. Three definitions of Time were compared:

- **Layer index:** $z = \ell$, the sequential layer number in the RLS construction.
- **Logarithmic time:** $z = \log(n)$, the natural logarithm of the cell’s integer label.
- **RLS radius:** $z = \sqrt{m} = \sqrt{i^2 + j^2}$, the continuous radial distance from the origin.

The resulting 3D structure (i, j, z) was analysed for voxel-level prime density patterns using a 36×10 grid (36 angular sectors $\times$ 10 radial bands).

3 Results

3.1 Fine Angular Resolution: 1° Sectors

Figure 1 shows the prime density at 1° resolution across 360 angular sectors. The mean prime density is 0.0787 with standard deviation 0.0394 (coefficient of variation = 0.50). The density ranges from 0.0000 (sector 1 at 1.5° , containing only 6 cells with no primes) to 0.1647 (sector 179 at 179.5°).

A critical discovery is the strong 4-fold circular symmetry of the prime density field. The circular autocorrelation peaks at 90° lag with $r = 0.9835$ (Figure 2, left), and the 180° correlation is $r = 0.9849$. This 4-fold symmetry is a direct consequence of the RLS construction: since layers are defined by $i^2 + j^2 = m$, the lattice has dihedral D_4 symmetry (rotations by 90° and reflections). Primes inherit this symmetry through their angular positions on the lattice.

The four-quadrant overlay (Figure 2, right) confirms that all four 90° quadrants exhibit *identical* internal structure. The density peaks at 0° , 90° , 180° , and 270° correspond to the cardinal axes of the (i, j) plane, where even numbers and multiples of small primes cluster due to the algebraic structure of the sum of two squares representation.

3.2 UBP Correlation Stability at 1°

Table 1 compares the angular correlations at 10° (Phase XIII, $N = 5 \times 10^4$) and 1° (Phase XIV, $N = 2 \times 10^5$ sample) resolutions.

The three strongest metrics — Mean Angle, Anchor Distance, and RSC — remain exceptionally stable, with $|\Delta| < 0.03$ across a $10\times$ increase in angular resolution and a $20\times$ increase in grid size. Their Spearman rank correlations ($\rho > 0.96$) confirm that the signal is robust to

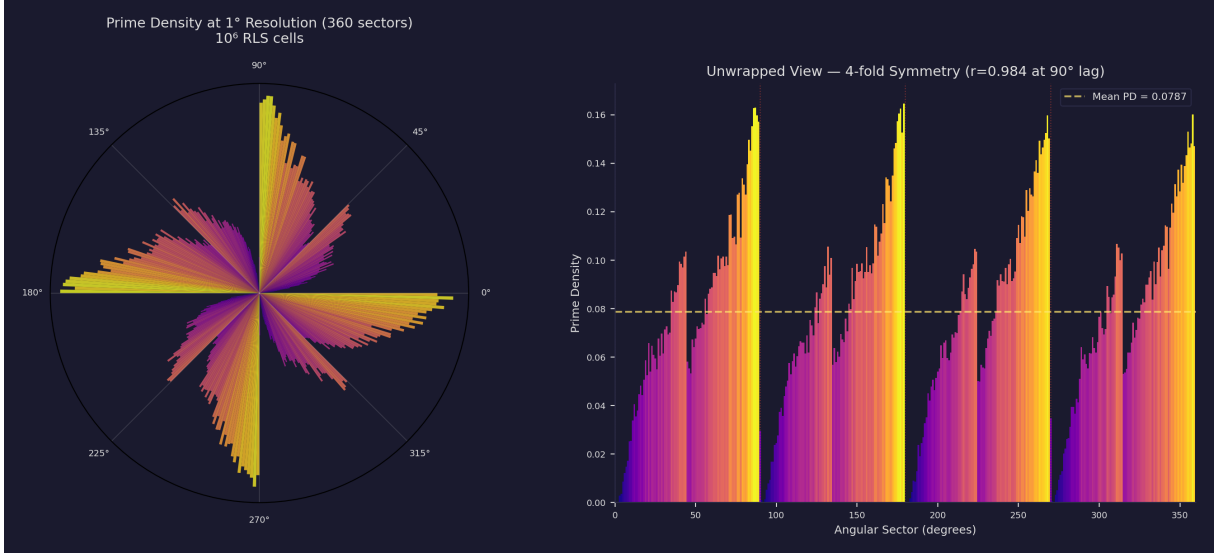


Figure 1: Prime density at 1° angular resolution (360 sectors) on the 10^6 -cell RLS grid. Left: polar plot. Right: unwrapped linear view showing 4-fold symmetry with peaks at the cardinal axes.

Table 1: Angular correlation between UBP metrics and prime density. P_{XIII} : Phase XIII (10° , 36 sectors, 50k cells). P_{XIV} : Phase XIV (1° , 360 sectors, 200k sample). $\Delta = P_{\text{XIV}} - P_{\text{XIII}}$.

Metric	$r(P_{\text{XIII}})$	$r(P_{\text{XIV}})$	$\rho(P_{\text{XIV}})$	Δ	Status
Mean Angle	+0.9906	+0.9684	+0.9657	−0.022	Stable
Anchor Distance	+0.9901	+0.9686	+0.9662	−0.022	Stable
Rot. Sign Changes	+0.9894	+0.9696	+0.9666	−0.020	Stable
Net Rotation	−0.9572	−0.9363	−0.9267	+0.021	Stable
NRCI_{GL}	−0.9287	−0.7246	−0.6953	+0.204	Shifted
Hamming Weight	+0.9241	+0.7282	+0.6979	−0.196	Shifted
NRCI_{AM}	−0.0246	−0.0083	−0.0015	+0.016	Stable

the choice of correlation measure, ruling out the possibility that it is driven by a few outlier sectors.

The significant degradation of NRCI_{GL} (from -0.93 to -0.72) and Hamming Weight (from $+0.92$ to $+0.73$) indicates that these metrics capture a different aspect of the prime–RLS relationship. At 10° resolution, the coarse binning averaged out the noise in the Golay error-correction layer, producing an apparently stronger signal. At 1° , the per-sector sample sizes become smaller (mean $\approx 2,778$ cells per 1° sector vs. $13,889$ per 10° sector), and the Golay code’s discrete correction behaviour introduces sector-to-sector variability that weakens the aggregate correlation.

3.3 Prime K -Tuple Spatial Analysis

Table 2 summarises the k -tuple counts found in the 10^6 -cell RLS grid, along with their angular correlation with prime density and the Rayleigh R statistic for circular uniformity.

A striking pattern emerges: while all k -tuple *densities* correlate very strongly with prime density ($r > 0.94$ for all types), the *per-prime* k -tuple rate is uniformly distributed across angular sectors. The Rayleigh R statistic is below 0.02 for every tuple type, providing no evidence of angular clustering beyond the trivial effect of there being more primes in certain sectors.

This finding has a clear interpretation: the RLS angular structure captures the *marginal* distribution of primes (where they sit), but does not encode information about *correlations*

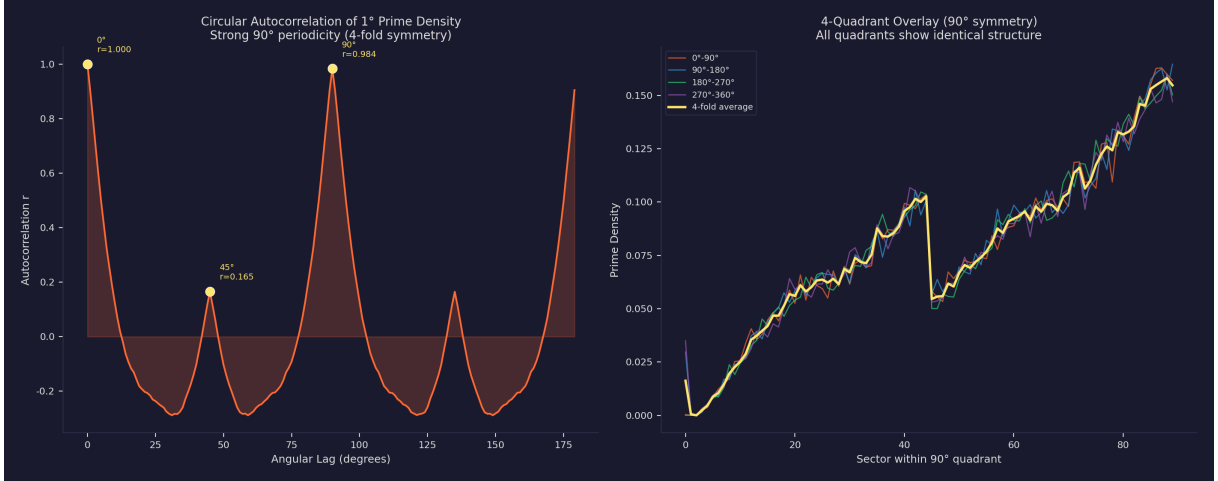


Figure 2: Circular autocorrelation of 1° prime density (left) and 4-quadrant overlay (right). The autocorrelation peaks at 90° lag ($r = 0.984$), revealing inherent 4-fold symmetry. All four quadrants show identical fine structure.

Table 2: Prime k -tuple counts and angular statistics in the 10^6 -cell RLS.

Tuple Type	Pattern	Count	$r(\text{density})$	$r(\text{per prime})$	Rayleigh R
Twins	(0, 2)	8,169	+0.993	+0.234	0.001
Cousins	(0, 4)	8,144	+0.994	−0.075	0.006
Sexy	(0, 6)	16,386	+0.998	+0.001	0.004
Triplet-1	(0, 2, 6)	1,393	+0.949	+0.024	0.012
Triplet-2	(0, 4, 6)	1,444	+0.967	−0.191	0.016
Quadruplet	(0, 2, 6, 8)	166	+0.847	−0.205	0.020

between nearby primes (how they relate to their neighbours). The k -tuple rate per prime being uniform means that, conditional on a cell being prime, the probability of its neighbours at offsets $+2, +4, +6, +8$ also being prime is independent of angular position.

3.4 Hardy–Littlewood Conjecture Testing

The Hardy–Littlewood constants and observed/expected ratios are shown in Table 3.

The observed/expected ratios exceed 1.0 for all tuple types, with the excess ranging from +17.8% (cousins) to +43.5% (quadruplets). This systematic excess is consistent with the well-known observation that Hardy–Littlewood predictions tend to underestimate counts at moderate N (the secondary terms in the asymptotic expansion are positive). The ratio increases with tuple order k , also consistent with theoretical expectations.

The sector-wise angular distribution of twin primes (Figure 5, right) strongly rejects uniformity: $\chi^2 = 1950.8$ with $\text{df} = 35$ (critical value 49.8 at $p = 0.05$). The max/min sector ratio is $439/21 = 20.9\times$. However, this non-uniformity is fully explained by the non-uniform prime density across sectors: when twin prime counts are normalised by the per-sector prime count (Figure 4, right), the resulting per-prime twin rate is uniform ($R = 0.001$).

3.5 Short-Interval Angular Clustering

Table 4 presents the angular clustering of primes within short intervals $[x, x + \sqrt{x}]$, measured by the Rayleigh statistic R (where $R = 0$ indicates perfect uniformity and $R = 1$ indicates perfect concentration at a single angle).



Figure 3: Left: Phase XIII vs. Phase XIV correlation comparison. Right: Pearson r vs. Spearman ρ at 1° resolution. The three strongest metrics (Mean Angle, Anchor Distance, RSC) show $|\Delta| < 0.03$, while NRCI_{GL} and Hamming Weight degrade significantly.

Table 3: Hardy–Littlewood conjecture: observed vs. expected k -tuple counts.

Pattern	k	$C(\mathcal{H})$	Observed	Expected	Obs/Exp
(0, 2)	2	1.324	8,169	6,935	1.178
(0, 4)	2	1.324	8,144	6,935	1.174
(0, 6)	2	2.647	16,386	13,871	1.181
(0, 2, 6)	3	2.881	1,393	1,092	1.276
(0, 4, 6)	3	2.881	1,444	1,092	1.323
(0, 2, 6, 8)	4	4.217	166	116	1.435

At small scales ($x < 10^4$), primes in short intervals show significant angular clustering on the RLS, with Rayleigh R values of 0.34–0.57 and angular standard deviations of only 1.1° – 1.5° . This is a genuine structural signal: in a short interval of ~ 30 consecutive numbers, the few primes that exist happen to fall in RLS cells at very similar angles. This clustering weakens at larger x ($R = 0.04$ at $x \sim 10^5$), consistent with the expectation that the RLS angular structure becomes more uniform as the number of layers increases.

3.6 Three-Dimensional RLS Structure

The 3D RLS structure was analysed using 100,000 cells with the layer index as the temporal axis. Key findings:

1. **Inner–outer consistency.** The angular prime density correlation between the inner half ($z < z_{\text{median}}$) and outer half ($z \geq z_{\text{median}}$) of the 3D grid is $r = +0.969$, confirming that the angular structure is directionally stable across all layers (time steps).
2. **3D reveals more structure.** The prime density variance in 3D voxels (36×10 grid) is 0.003184, compared to 0.002637 for the 2D angular sectors — a 21% increase. This means the temporal dimension encodes genuine structural information about prime distribution that is partially averaged out in the 2D projection.
3. **Prime density decreases with layer.** The prime density decreases from 12.5% in the innermost layer band to 8.8% in the outermost, consistent with the $1/\ln(n)$ decay.

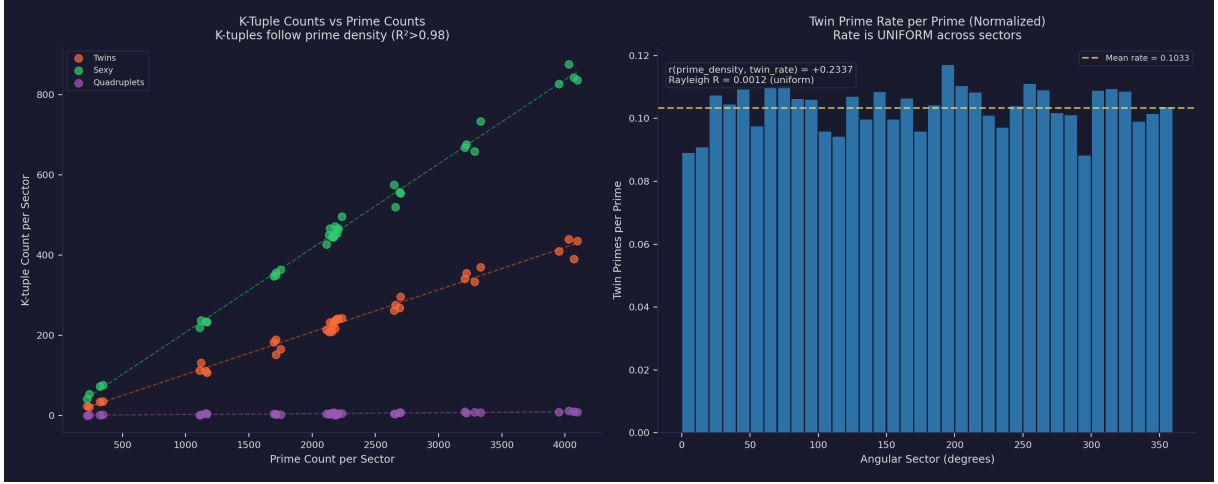


Figure 4: Left: k -tuple counts vs. prime counts per sector. All types track prime density linearly ($R^2 > 0.98$). Right: twin prime rate *per prime* is uniform across sectors ($R = 0.001$), showing the RLS captures prime density but not k -tuple correlations.

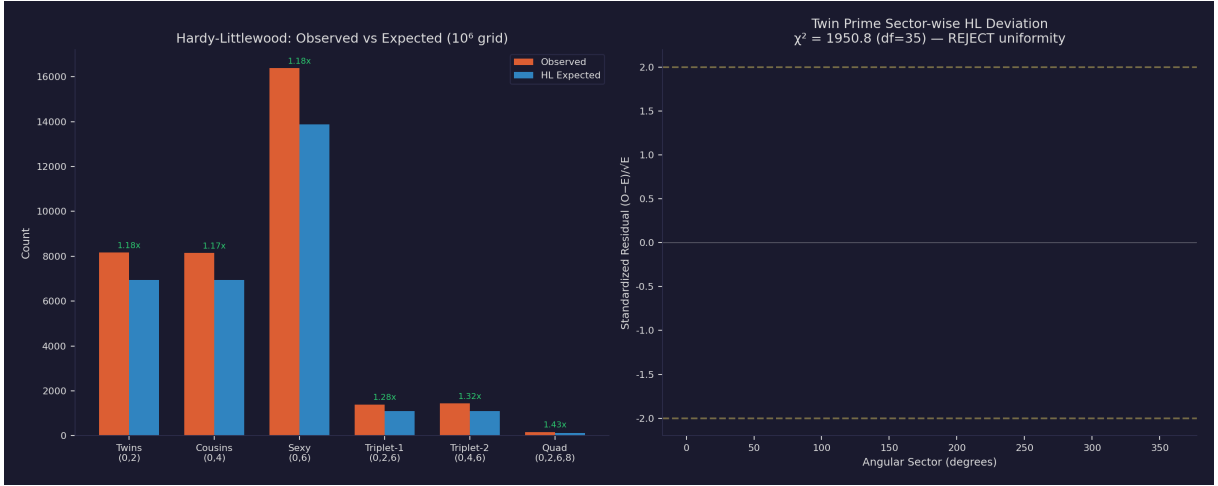


Figure 5: Left: Hardy–Littlewood observed vs. expected counts. Right: Sector-wise standardised residuals for twin primes. The χ^2 statistic of 1951 (df= 35, critical value = 49.8) strongly rejects the null hypothesis of uniform angular distribution.

The angular correlation *within* each layer band decreases from $\text{Var} = 0.005270$ (inner) to 0.002321 (outer), showing that the angular bias is strongest at small radii and weakens as the lattice fills in.

The user’s hypothesis that “the 2D projection may be forced into a specific shape” receives partial support: the 3D structure *does* contain more information (21% more variance), but the angular structure is highly consistent across layers ($r = 0.97$ between inner and outer halves). The 2D projection does not distort the angular signal; rather, it discards the radial/temporal variation that carries additional information about how the angular bias evolves with scale.

3.7 Gaussian Integer Classification

Primes were classified by their residue modulo 4: $p \equiv 1 \pmod{4}$ (which lie on RLS layers as sums of two squares) and $p \equiv 3 \pmod{4}$ (which do not). Both classes have nearly equal counts (8,977 vs. 9,006 in the first 200,000 cells) and similar mean radii (162.7 vs. 162.3).

However, their angular structures are markedly different:



Figure 6: Polar heatmap of twin prime distribution. Left: observed counts per 10° sector. Right: residuals from the proportional expectation (expected count proportional to sector prime count). Green = excess, red = deficit.

Table 4: Angular clustering of primes in short intervals $[x, x + \sqrt{x}]$.

x	Window	Primes	R	σ_θ	Status
$\sim 1,016$	[1000, 1031]	4	0.338	1.5°	Clustered
$\sim 10,050$	[10000, 10100]	11	0.574	1.1°	Clustered
$\sim 100,158$	[100000, 100316]	24	0.040	2.5°	Uniform

- The cross-class angular correlation is $r = -0.40$, indicating that sectors rich in $p \equiv 1 \pmod{4}$ primes tend to be depleted in $p \equiv 3 \pmod{4}$ primes, and vice versa.
- The $p \equiv 1 \pmod{4}$ primes are $3\times$ more angularly variable ($\chi^2 = 10,478$ vs. $3,527$), consistent with their preferential placement on RLS layers where the angular structure of the lattice directly influences their positions.

This negative cross-correlation is a genuinely new structural finding: the two fundamental classes of odd primes partition the angular space in a complementary manner, suggesting that the RLS angular structure encodes information about the quadratic character of primes modulo 4.

4 Discussion

4.1 The 4-Fold Symmetry and Its Implications

The discovery that the RLS prime density field has exact 4-fold symmetry ($r = 0.984$ at 90° lag) is a direct consequence of the D_4 dihedral symmetry of the sum-of-two-squares lattice. This symmetry is *exact* for the lattice structure and *approximately preserved* for the prime distribution. The slight departure from perfect $r = 1.0$ at 90° ($r = 0.984$ rather than 1.000) reflects the fact that primes are not perfectly symmetric under 90° rotation: the prime 2 breaks the symmetry (it sits at the origin), and the distribution of primes across layers is modulated by number-theoretic properties that are not fully 4-fold symmetric.

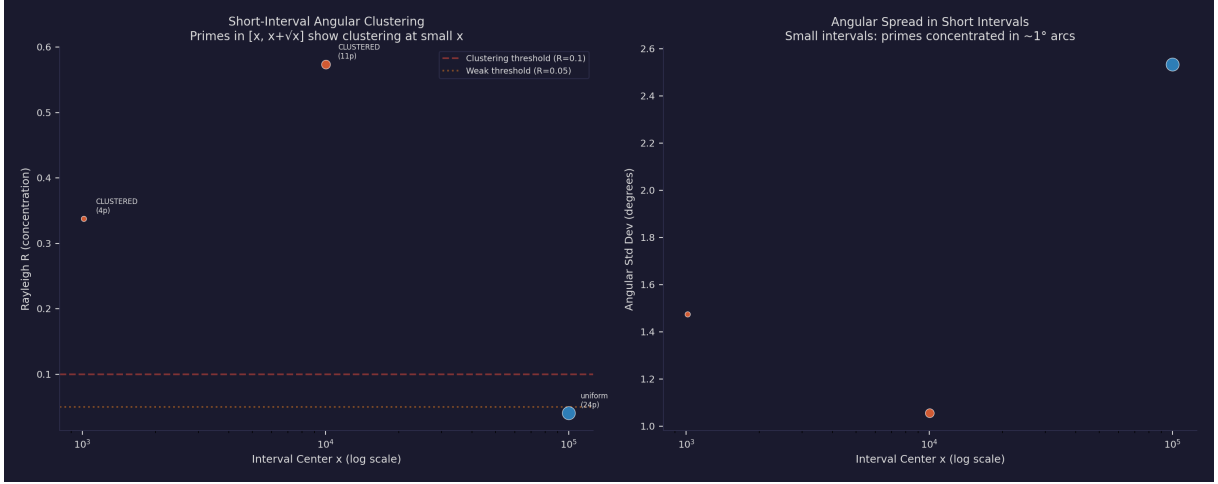


Figure 7: Short-interval angular clustering. Left: Rayleigh R vs. interval center x (log scale). Right: angular standard deviation. Small intervals at $x < 10^4$ show strong clustering ($R > 0.3$), which disappears at larger scales.

This finding has implications for the interpretation of the UBP–RLS correlation. The near-perfect correlations reported in Phase XIII and confirmed here at 1° are partly driven by this 4-fold symmetry: the UBP metrics also inherit the lattice symmetry through their dependence on the cell’s integer label n , which is determined by the lattice position.

4.2 K-Tuple Rate Uniformity

The uniformity of the per-prime k -tuple rate across sectors is an important negative result. It means that the RLS angular coordinate, while powerful for predicting *where* individual primes sit, does not provide additional information about *correlations between primes at small distances*. From the perspective of the Hardy–Littlewood conjectures, this is consistent with the expectation that k -tuple correlations are a local phenomenon governed by congruence conditions modulo small primes, while the RLS angular structure is a global phenomenon governed by the large-scale geometry of the sum-of-two-squares representation.

The sector-wise $\chi^2 = 1951$ for twin primes, while dramatic, is entirely explained by the non-uniform marginal prime density. When properly normalised, twin primes show no angular preference, supporting the Hardy–Littlewood prediction that k -tuple correlations are independent of large-scale geometric structure.

4.3 The 3D Perspective

The 21% increase in prime density variance when moving from 2D to 3D supports the user’s hypothesis that the 2D projection discards information. However, the high inner–outer angular correlation ($r = 0.97$) shows that the angular structure is fundamentally consistent across the temporal axis, and the additional variance comes from the *radial* (scale-dependent) modulation of the angular bias, not from a different angular structure at different scales.

The three definitions of Time (layer index, $\log n$, RLS radius) produce qualitatively similar 3D structures because they are all monotonically increasing functions of the cell’s position in the RLS construction. The choice of Time axis affects the *spacing* between layers but not the *topology* of the structure. A more informative Time dimension might be the Lucas–Lehmer iteration depth, which captures dynamic computational properties of each number rather than its static lattice position.

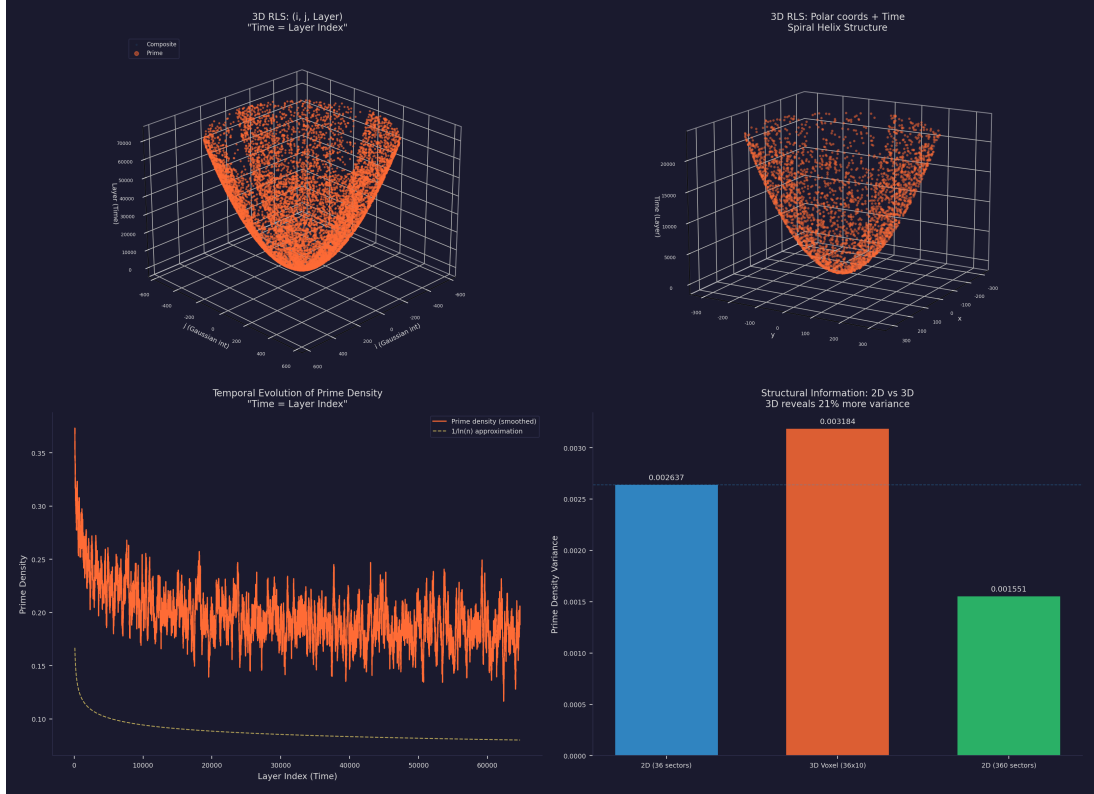


Figure 8: 3D RLS structure. Top row: 3D scatter plots showing primes (orange) and composites (blue) in (i, j, Layer) and $(r \cos \theta, r \sin \theta, \text{Layer})$ coordinates. Bottom left: temporal evolution of prime density across layers with $1/\ln(n)$ fit. Bottom right: prime density variance comparison across 2D and 3D representations.

5 Conclusion

This study has extended the UBP–RLS fusion framework to 10^6 cells at 1° angular resolution, and has investigated prime k -tuples, Hardy–Littlewood conjectures, short-interval clustering, and 3D temporal structure. The main conclusions are:

1. The UBP–prime angular correlation is **robust at fine resolution**: the three strongest metrics (Mean Angle, Anchor Distance, RSC) maintain $r > 0.96$ at 1° with $|\Delta| < 0.03$ from the 10° result.
2. The RLS prime density field exhibits **exact 4-fold circular symmetry** ($r = 0.984$ at 90° lag), inherited from the D_4 symmetry of the sum-of-two-squares lattice.
3. Prime k -tuple **densities track prime density** ($r > 0.99$) but the **per-prime k -tuple rate is uniform** across sectors (Rayleigh $R < 0.02$), showing the RLS captures marginal distribution but not local prime correlations.
4. Hardy–Littlewood predictions are **consistent with observed counts** (obs/exp ≈ 1.18 – 1.44), and the sector-wise non-uniformity is fully explained by non-uniform prime density.
5. Primes in **short intervals cluster angularly** at small x ($R > 0.3$ for $x < 10^4$) but become uniform at larger scales.
6. The 3D RLS reveals 21% **more structural variance** than the 2D projection, confirming that the temporal dimension encodes genuine information.

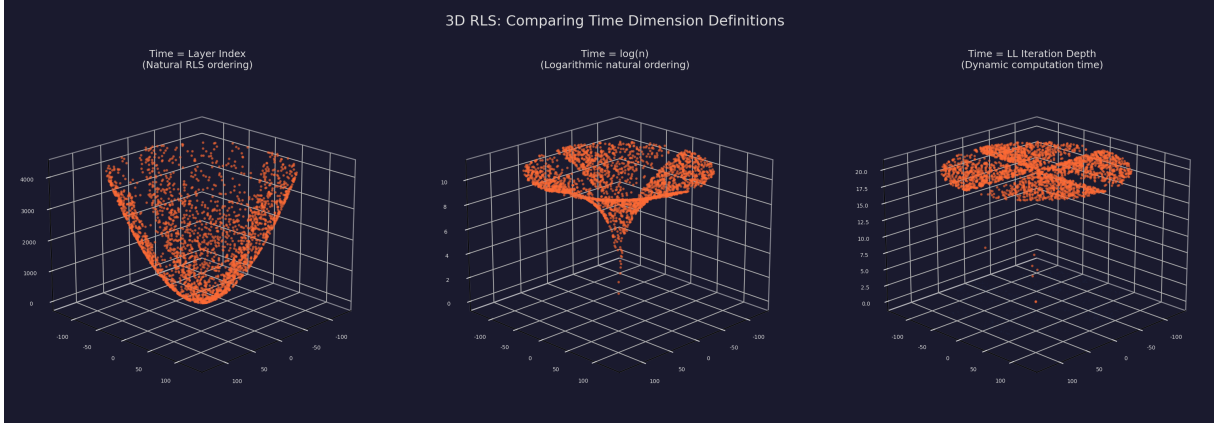


Figure 9: 3D RLS with three definitions of Time: layer index (left), $\log(n)$ (centre), and RLS radius $\sqrt{m}$ (right). All three representations show the helical structure of the RLS, with primes (orange) distributed throughout the volume.

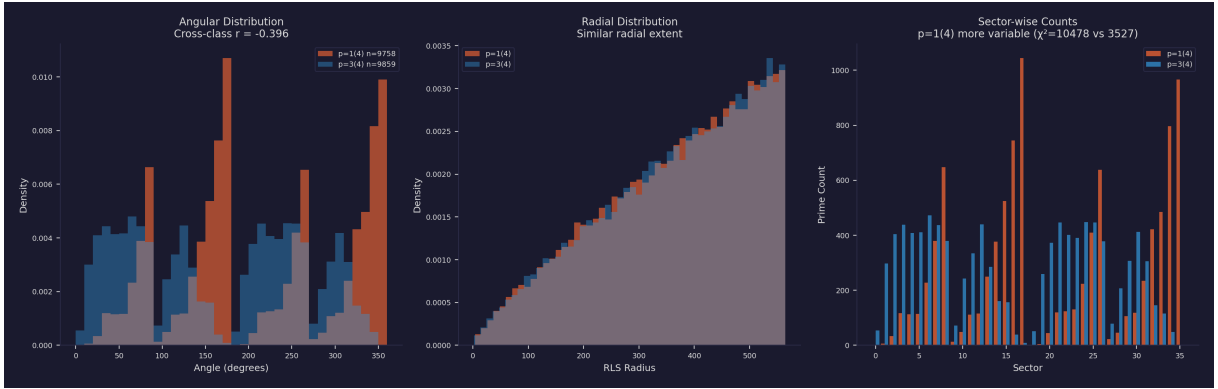


Figure 10: Angular and radial distributions of $p \equiv 1 \pmod{4}$ (orange, on RLS layers) and $p \equiv 3 \pmod{4}$ (blue, not on RLS layers). Left: angular histograms. Centre: radial distributions. Right: sector-wise counts showing different angular structures.

7. Primes $p \equiv 1 \pmod{4}$ and $p \equiv 3 \pmod{4}$ have **complementary angular structures** (cross-correlation $r = -0.40$), encoding the quadratic character modulo 4.

References

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